The response of the colon to eating.

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Much evidence indicates that there is an increased motor activity in the colon from eating in several species of animals. Though some of this effect may be cephalic in origin, the greater part of the response results from the arrival of food in the stomach and proximal intestine. Chemoreceptor stimulation appears to be more important than mechanoreceptor stimulation in bringing about this effect. The means by which this effect comes about could be either hormonal or neural. Several polypeptide hormones released from the proximal gut by eating are candidates. Neural pathways through both parasympathetic and sympathetic systems could be responsible. The exact nature of the change in colonic motility that is produced is unknown. It could involve changes in the pacemakers for colonic contractions (the electrical slow waves of the colon and the migrating spike burst of the colon), changes in the excitability of the colonic musculature or changes in colonic mucosal function.